

$$\tau_p \frac{dy}{dt} + y = K_p f(t) \quad - (1)$$

$y(0) = 0; \quad f(0) = 0$

Apply Laplace transform: -

~~$$\tau_p \{sY(s)\} + Y(s) = K_p$$~~

$$\tau_p \{sY(s)\} + Y(s) = K_p F(s)$$

$$Y(s) \{\tau_p s + 1\} = K_p F(s) \quad - (2)$$

$$Y(s) = \frac{K_p F(s)}{\tau_p s + 1}$$

$$\frac{Y(s)}{F(s)} = \frac{K_p}{\tau_p s + 1}$$

= Transfer function